

Effect of Fatty Liver on Hepatic Gluconeogenesis in Periparturient Dairy Cows

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ABSTRACT

The purpose of this study was to compare the hepatic enzyme activities of gluconeogenesis between control cows and experimental cows that had been overfed during the dry period to induce fatty liver postpartum. Blood and liver samples were collected 1 wk before and 0.5, 1, 2, and 3 wk after parturition. Before parturition, neither the serum nonesterified fatty acid nor the liver triacylglycerol concentration differed between the two groups. After parturition, these variables were higher in experimental cows than in control cows. Liver glycogen was higher at 1 wk before parturition in experimental cows; sharply decreased after parturition in both groups; and, at 1 wk after parturition, was lower in experimental cows than in control cows. In the liver, activities of phosphoenolpyruvate carboxykinase were significantly lower at 1 wk before and at 0.5 and 2 wk after parturition in experimental cows; in addition, the activities tended to be lower at 1 wk after parturition. Activities of fructose 1,6-bisphosphatase tended to be lower, but activities of glucose 6-phosphatase tended to be higher, at 0.5 wk after parturition in experimental cows than in control cows. Our results suggest that, in fatty infiltrated liver, the rate of gluconeogenesis is not optimal, which results in prolongation of lipolysis, particularly during the first weeks after parturition.

(**Key words:** dairy cows, gluconeogenesis, fatty liver)

Abbreviation key: **FBPase** = fructose 1,6-bisphosphatase, **G6-Pase** = glucose 6-phosphatase, **NEB** = negative energy balance, **PEPCK** = phosphoenolpyruvate carboxykinase, **TAG** = triacylglycerol.

INTRODUCTION

Although ruminants absorb considerable amounts of VFA, mainly acetate, propionate, and butyrate, from the forestomach, they absorb little glucose from the intestinal tract (3). Thus, circulating glucose in ruminants is derived from gluconeogenesis, primarily in the liver (16), which depends on the availability of glucogenic precursors. In fed ruminants, propionate is the major source of gluconeogenesis, but glycerol and glucogenic AA are the predominant sources in ruminants deprived of feed (3). A relative shortage of energy occurs in cows during early lactation when the energy requirements are much greater than the energy intake from the feed, a condition known as negative energy balance (**NEB**). Overconditioning or excess weight at calving is a predisposing factor for the cow to enter a more severe NEB after parturition and to develop fatty liver (15).

Data are limited as to the effect on gluconeogenesis when cows experience both severe NEB and increased hepatic uptake of fatty acids, which result in the development of fatty liver. Mills et al. (8) demonstrated that ketotic cows accumulated greater amounts of triacylglycerols (**TAG**) in the liver than did nonketotic cows. In addition, ketotic cows had a decreased hepatic gluconeogenic capacity (9).

Therefore the objective of this study was to collect more information about the activities of three key enzymes for hepatic gluconeogenesis [i.e., phosphoenolpyruvate carboxykinase (**PEPCK**), fructose 1,6-bisphosphatase (**FBPase**), and glucose 6-phosphatase (**G6-Pase**)] in periparturient dairy cows that had been overfed during the dry period to induce fatty liver postpartum and to compare these activities with the activities of the same three enzymes in control cows.

MATERIALS AND METHODS

Cows and Feeding

We used 13 pregnant, multiparous Holstein-Friesian cows (age, 3.5 ± 1.1 yr; mean 305-d milk

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production, 7874 ± 1044 kg; and mean BW, 638 ± 85 kg at the start of the experiment; $\bar{X} \pm SD$). The cows were randomly allocated to two groups; 6 cows were assigned to the control group, and 7 cows were assigned to the experimental group, which was administered a feeding regimen to induce fatty liver (15). The cows were housed in a stanchion barn with wheat straw as bedding material. The cows were fed individually and had free access to tap water.

During the dry period (10 wk), the control cows were daily fed a diet consisting of 1 kg of corn silage (DM basis), 3.5 kg of grass silage (DM basis), 0.5 kg of extracted soybean meal, 0.5 kg of sugar beet pulp pellets, and 1 kg of rapeseed meal. Wheat straw was offered for ad libitum intake. The experimental cows were daily fed a diet consisting of 2 kg of corn silage, 1 kg of extracted soybean meal, 2 kg of sugar beet pulp pellets, and 3 kg of rapeseed meal. Grass silage was offered for ad libitum consumption. With these diets, the control cows received approximately 44 MJ/d of NE_L ; the test cows received 96 MJ/d of NE_L .

After parturition, the cows in both groups were daily fed the same lactation diet, which was composed of 5 kg of commercial concentrate, 2 kg of corn silage, 1 kg of extracted soybean meal, 2 kg of sugar beet pulp pellets, and 3 kg of rapeseed meal. Grass silage was offered for ad libitum consumption. This diet delivered approximately 164 MJ/d of NE_L . To induce a deeper NEB, the experimental cows had been deprived of feed for 8 h immediately after delivery.

Cows were weighed at regular intervals before and after parturition, and milk production was recorded daily.

Sampling Procedures

Blood and liver samples were collected at 1 wk before expected parturition and at 0.5, 1, 2, and 3 wk after parturition. Blood was collected from the jugular vein in evacuated tubes (Vaccuette® vacuum tubes; Greiner Labortechnik, Alphen a/d Rijn, The Netherlands) containing paraoxon (Diethyl 4-nitrophenyl phosphate; Aldrich Chemical Co., Inc., Milwaukee, WI) for the determination of NEFA. All samples were stored at -20°C until analyses.

Liver samples were obtained by biopsy using the method described by Van den Top et al. (15). For TAG and liver enzyme determination, the liver samples were placed in physiological saline and carried to the laboratory in a thermostatic (0 to 4°C) container.

For glycogen determination, the samples were placed in 0.5 ml of 20% KOH solution.

Preparation of Liver Samples for the Determination of Enzyme Activity

In the laboratory, the liver samples were homogenized with the IKA-Ultra Turrax® T5-FU tissue homogenizer (Janke & Knukel GmbH & Co. KG, Staufen, Germany) in 9 vol of a buffer (pH 8.0) containing 25 mM HEPES and 5 mM β -mercaptoethanol. The homogenate was centrifuged at $48,000 \times g$ for 30 min. The supernatant was filtered and stored at -80°C until analyses.

Assay Procedures

Serum NEFA concentrations were measured enzymatically with a commercially available kit (kit number 442640; Beckman Instruments B. V., Mijdrecht, The Netherlands). Liver TAG concentrations were assayed with the use of a commercial kit (kit number 337-A; Sigma Chemical Co., St. Louis, MO). Liver glycogen concentrations were determined as described by Van den Top et al. (14).

The activity of PEPCK (EC 4.1.1.32) was assayed by incorporation of $[^{14}\text{C}]\text{NaHCO}_3$ (2). The activity of FBPase (EC 3.1.3.11) was determined spectrophotometrically (12). The activity of G6-Pase (EC 3.1.3.9) was determined by monitoring the production of radiolabeled glucose from $[\text{U-}^{14}\text{C}]\text{-G6-phosphate}$ (11).

Statistical Analyses

In the control group, 1 cow had twin calves, and 1 cow aborted her calf; therefore, data for these 2 cows were excluded from the analyses. All analyses were performed using a computer program from the Statistical Package for the Social Sciences (10). Data were tested for normal distribution using the Komogorov-Smirnov test. Normally distributed data were subjected to analysis of variance using dietary treatment as the fixed main effect and sampling day as a repeated measure. Data from the control and the experimental groups were compared using Student's *t* test or, when data were not normally distributed, using the Mann-Whitney U test. When the data had a normal distribution, homogeneity of variances was verified using Levene's test. Two-sided statistical significance was determined at $P \leq 0.05$.

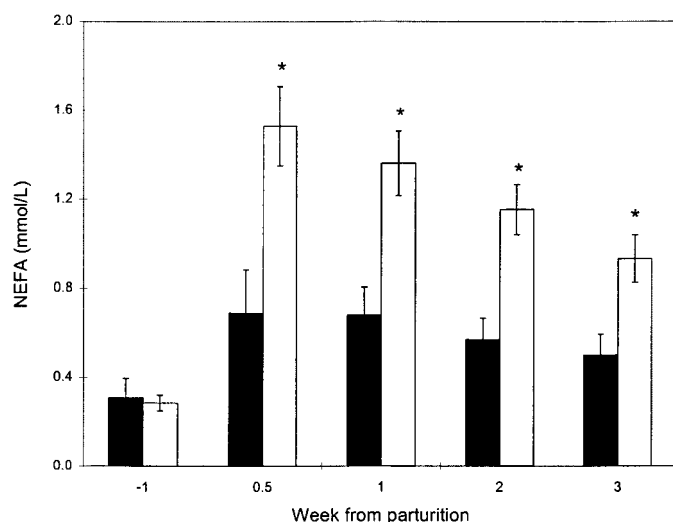


Figure 1. Serum NEFA concentrations ($\bar{X} \pm \text{SEM}$) measured before and after parturition, were compared between control cows ($n = 4$; ■) and experimental cows ($n = 7$; □). Asterisks indicate a difference ($P < 0.05$) in the mean concentration of serum NEFA between the two groups.

RESULTS

Performance

During the dry period, experimental cows consumed more calories than did control cows (13), and, as a result, mean BW gain was greater for experimental cows than for control cows [124.0 ± 12.1 vs. 67.8 ± 12.8 kg ($\bar{X} \pm \text{SEM}$); $P = 0.02$]. After parturition, mean BW loss (i.e., the difference between BW measured immediately after parturition and that 5 wk later) was greater for experimental cows than for control cows (60.5 ± 4.0 vs. 16.3 ± 3.2 kg; $P < 0.001$). During the 1st through the 5th wk of lactation, mean milk production was 36.2 kg/d (SD = 3.4) and 36.4 kg/d (SD = 2.7) for experimental cows and control cows, respectively, and did not significantly differ between the two groups.

NEFA

Mean concentrations of serum NEFA are presented in Figure 1. The concentrations did not differ between the two groups at 1 wk before parturition but rose sharply after parturition in both groups. Compared with the concentration at 1 wk before parturition, the concentration at 0.5 wk after parturition increased 446% in the experimental group but increased only 123% in the control group. Moreover, the concentration of serum NEFA was significantly higher at 0.5, 1,

2, and 3 wk after parturition in experimental cows than in control cows.

TAG and Glycogen

Mean concentrations of liver TAG and glycogen are presented in Figure 2. At 1 wk before parturition, the concentration of TAG (Figure 2a) did not differ between the two groups. However, the TAG concentration was significantly higher at 0.5, 1, 2, and 3 wk after parturition in experimental cows than in control cows. When compared with the value at 1 wk before parturition, the concentration at 0.5 wk after parturi-

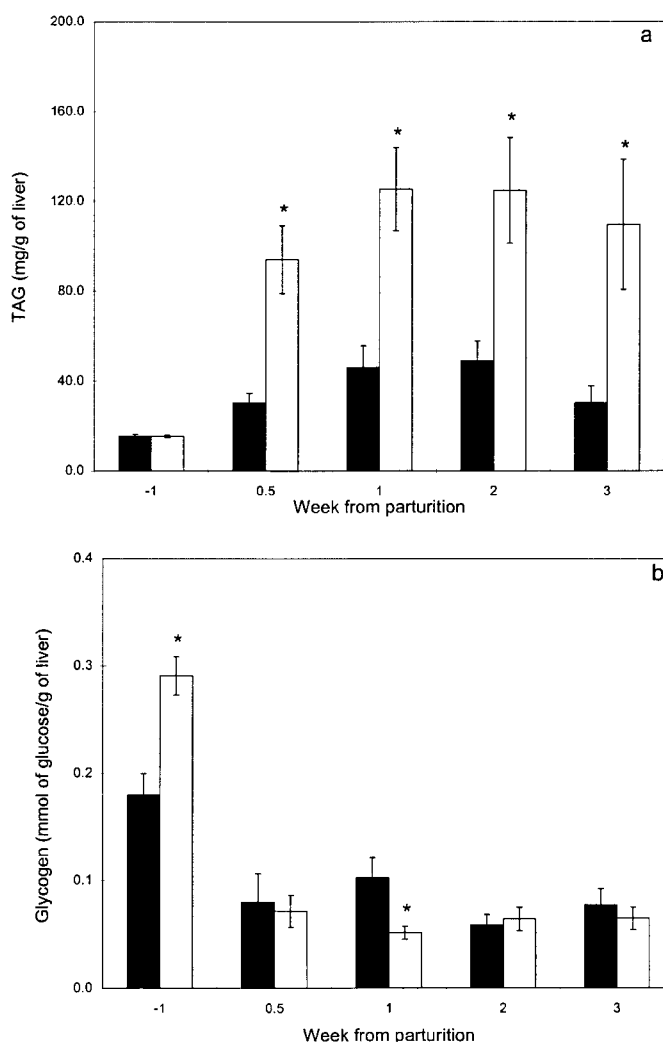


Figure 2. Triacylglycerol (TAG; a) and glycogen (b) concentrations ($\bar{X} \pm \text{SEM}$) in the liver, measured before and after parturition, were compared between control cows ($n = 4$; ■) and experimental cows ($n = 7$; □). Asterisks indicate a difference ($P < 0.05$) in the mean concentration of TAG or glycogen in the liver between the two groups.

tion increased 514% in experimental cows but increased only 97% in control cows. In both groups, the highest concentration was observed between 1 to 2 wk after parturition. The glycogen concentration (Figure 2b) was significantly higher at 1 wk before parturition in experimental cows than in control cows. After parturition, the concentration sharply decreased in both groups and was significantly lower at 1 wk after parturition in experimental cows than in control cows.

Gluconeogenic Enzyme Activities

Gluconeogenic enzyme activities in the liver are demonstrated in Figure 3. The activities of PEPCK were significantly lower 1 wk before and 0.5 and 2 wk after parturition in experimental cows than in control cows, and the same activities tended to be lower at 1 wk after parturition in experimental cows ($P < 0.1$) (Figure 3a). At 3 wk after parturition, the activities of this enzyme did not differ between the two groups.

At 1 wk before parturition, the activities of FBPase did not differ significantly between the two groups (Figure 3b); however, the activities tended ($P < 0.1$) to be lower for experimental cows. The same was true at 0.5 wk after parturition. From 1 to 3 wk after parturition, there was no difference in the activities of this enzyme between the two groups.

Before parturition, the activities of G6-Pase did not differ between the two groups (Figure 3c). The activities of this enzyme increased at 0.5 wk after parturition in both groups compared with the value at 1 wk before parturition but tended ($P < 0.1$) to be higher at 0.5 wk after parturition in experimental cows than in control cows. From 1 to 3 wk after parturition, the activities of this enzyme were not different between groups.

DISCUSSION

Overfeeding during the dry period leads to a deposition of body fat, which is corroborated by the observation that, in our experiment, cows that were overfed gained more BW than did control cows. Furthermore, cows that were overfed during the dry period increased mobilization of body fat after parturition more substantially than did control cows, probably because they went into a deeper NEB. This hypothesis is supported by a significantly higher serum NEFA concentration in experimental cows, a phenomenon that was also described earlier (13). An increased concentration of serum NEFA is associated with an increase in TAG synthesis in the liver (4, 13); this was also confirmed by our study, which

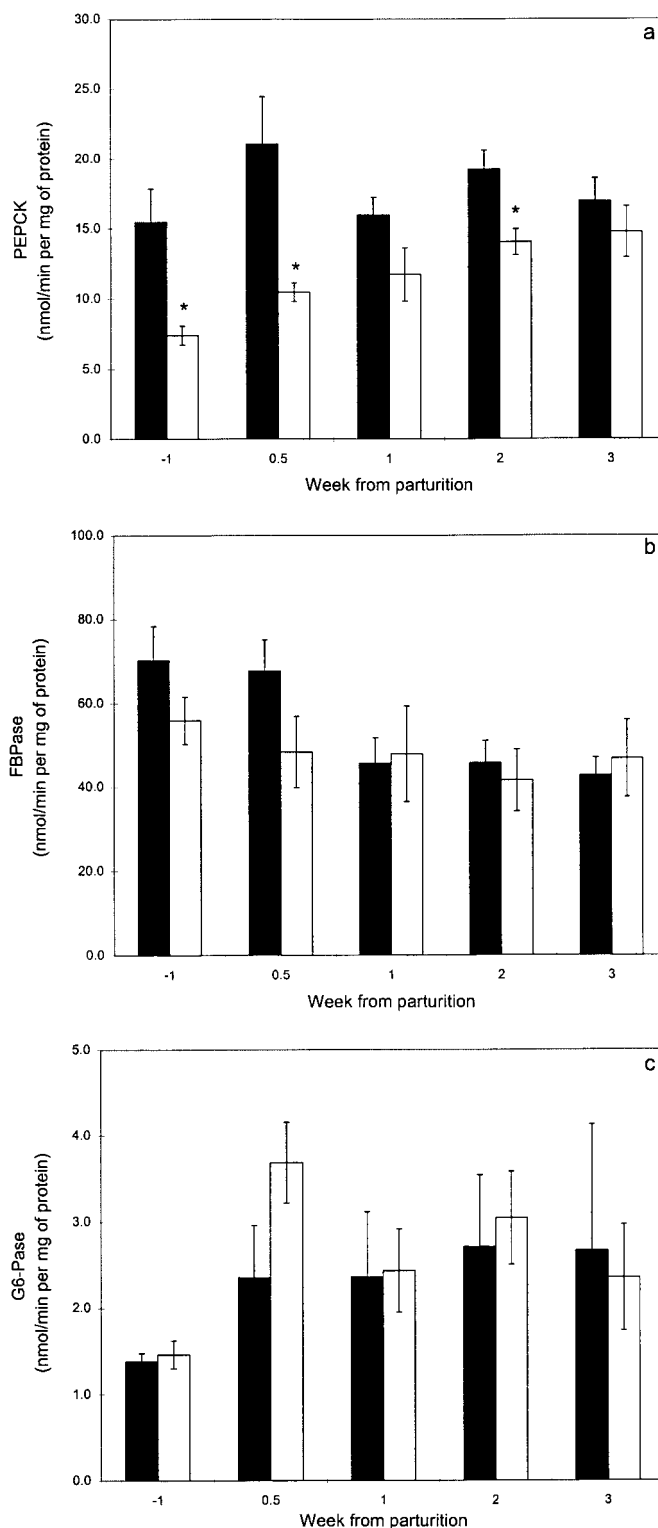


Figure 3. Activities ($\bar{X} \pm \text{SEM}$) of a) phosphoenolpyruvate carboxykinase (PEPCK), b) fructose, 1,6-bisphosphatase (FBPase), and c) glucose 6-phosphatase (G6-Pase) in the liver, measured before and after parturition, were compared between control cows ($n = 4$; ■) and experimental cows ($n = 7$; □). Asterisks indicate a difference ($P < 0.05$) in the mean activity of the enzymes between the two groups.

showed that experimental cows accumulated greater amounts of TAG in the liver after parturition.

Our study demonstrated that the activities of hepatic gluconeogenic enzymes, particularly PEPCK, in experimental cows were lower than those in control cows. Phosphoenolpyruvate carboxykinase is a crucial enzyme that converts cytosolic oxaloacetate to phosphoenolpyruvate in the gluconeogenic pathway. In fed ruminants, oxaloacetate is mostly derived from propionate (7). In our study, we observed that, already before parturition, the activities of PEPCK in the liver of experimental cows were lower than those in control cows. The possible reason underlying this finding is that a high energy diet results in the direct absorption of glucose from the forestomach as was demonstrated by Huntington and Prior (6). This amount of glucose was probably sufficient for the requirements of the cows; therefore, hepatic gluconeogenesis was minimized in experimental cows during that period. In contrast, the control cows were fed a low energy diet in the dry period, and, as a result, the glucose demand of these cows was dependent largely on gluconeogenesis, primarily in the liver (5). The livers of experimental cows were less adapted to gluconeogenesis during the dry period than were the livers of control cows.

Cows in both groups were fed the same diet during lactation; therefore, circulatory glucose was derived both from direct absorption, which is dependent on feed intake, and gluconeogenesis, which is dependent on glucogenic precursors. Experimental cows had been deprived of feed for 8 h from the onset of parturition; therefore, the high energy demands were compensated by lipolysis as indicated by a significant increase in serum NEFA concentration. Lipolysis in adipose tissue also releases glycerol, which is a prominent substrate for hepatic gluconeogenesis; this may imply that greater lipolysis results in a higher concentration of glycerol in the blood. Once glycerol enters the liver, it can be introduced into the gluconeogenic pathway via dihydroxyacetone phosphate, which can be converted to fructose 1, 6-bisphosphate, the substrate of FBPase. Although experimental cows had a greater lipolytic rate after parturition, we found that they had similar or lower activities of FBPase than did control cows. Because a rapid increase in NEFA in the liver results in a rapid increase of both synthesis and accumulation of TAG (4), we suggest that this fatty infiltration affects the ability of the liver to synthesize glucose from glucogenic precursors. Recently, Cardóniga-Valiño et al. (1) demonstrated *in vitro* that liver cells incubated with oleate had a lower rate of propionate conversion

to glucose than did cells without oleate.

Unlike PEPCK, the activities of G6-Pase increased after parturition. This enzyme may respond to the sudden need for glucose by converting glucose 6-phosphate to glucose. Available substrate for G6-Pase is not only from glucogenic precursors but also from the hydrolysis of liver glycogen. In the present study, the activity of this enzyme was higher in experimental cows at 0.5 wk after parturition, possibly because those cows had a significantly higher glycogen store in the liver before parturition, which might have provided more substrate for this enzyme in that period.

In conclusion, the hepatic gluconeogenic capacity in experimental cows is impaired. Because of the high concentrate diet fed during the dry period, the livers of experimental cows were less adapted to gluconeogenesis before parturition. After parturition, lower activities of PEPCK in experimental cows indicated an inadequate production of glucose. This particular situation promotes an ongoing mobilization of fatty acids from adipose tissue. As a consequence, the livers were continuously confronted with high concentrations of NEFA, which resulted in a more severe hepatic lipidosis. More research is necessary to quantify the impact of inadequate gluconeogenesis as affected by fatty infiltration of the liver for the recovery of energy balance in the periparturient cow.

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REFERENCES

- 1 Cardóniga-Valiño, C., R. R. Grummer, L. E. Armentano, S. S. Donkin, and S. J. Bertrics. 1997. Effects of fatty acids and hormones on fatty acid metabolism and gluconeogenesis in bovine hepatocytes. *J. Dairy Sci.* 80:646-656.
- 2 Duff, D. A., and K. Snell. 1982. Limitations of commonly used spectrophotometric assay methods for phosphoenolpyruvate carboxykinase activity in crude extracts of muscle. *Biochem. J.* 206:147-152.
- 3 Herdt, T. H. 1988. Metabolic diseases of ruminant livestock: fuel homeostasis in the ruminant. *Vet. Clin. North Am. Food Anim. Pract.* 4:213-231.
- 4 Herdt, T. H., T. Wensing, H. P. Haagsman, L.M.G. van Golde, and H. J. Breukink. 1988. Hepatic triacylglycerol synthesis during a period of fatty liver development in sheep. *J. Anim. Sci.* 66:1997-2013.
- 5 Huntington, G. B. 1997. Starch utilization by ruminants: from basic to the bulk. *J. Anim. Sci.* 75:852-867.
- 6 Huntington, G. B., and R. L. Prior. 1983. Digestion and absorption of nutrients by beef heifers fed a high concentrate diet. *J. Nutr.* 117:2280-2288.
- 7 Lomax, M. A., and G. D. Baird. 1983. Blood flow and nutrient exchange across the liver and gut of the dairy cows. Effects of lactation and fasting. *Br. J. Nutr.* 49:481-496.

- 8 Mills, S. E., D. C. Beitz, and J. W. Young. 1986. Characterization of metabolic changes during a protocol for inducing lactation ketosis in dairy cows. *J. Dairy Sci.* 69:352–361.
- 9 Mills, S. E., D. C. Beitz, and J. W. Young. 1986. Evidence for impaired metabolism in liver during induced lactation ketosis of dairy cows. *J. Dairy Sci.* 69:362–370.
- 10 SPSS Advance Statistic™. Version 6.1. 1994. Statistical Package Soc. Sci., Inc., Chicago, IL.
- 11 Tielens, A.G.M., J. M. Van den Heuvel, M.G.J. Schmitz, and M.J.H. Geelen. 1993. Effects of chronic benfluorex treatment on the activities of key enzymes of hepatic carbohydrate metabolism in old Sprague-Dawley rats. *Biochem. Pharmacol.* 46:1539–1544.
- 12 Ulm, E. H., B. M. Pogell, M. M. deMaine, C. B. Libby, and S. J. Benkovic. 1975. Fructose-1,6-bisphosphate from rabbit liver. Pages 369–374 *in* *Methods in Enzymology*. Vol. 42. W. A. Wood, ed. Acad. Press, New York, NY.
- 13 Van den Top, A. M., M.J.H. Geelen, T. Wensing, G. H. Wentink, A. T. van't Klooster, and A. C. Beynen. 1996. Higher postpartum hepatic triacylglycerol concentrations in dairy cows with free versus restricted access to feed during the dry period are associated with lower activities of hepatic glycerolphosphate acyltransferase. *J. Nutr.* 126:76–85.
- 14 Van den Top, A. M., T. Wensing, and A. C. Beynen. 1994. The influence of calcium palmitate and oleate feeding on hepatic lipid metabolism in dry goats. *J. Anim. Physiol. Anim. Nutr.* 72: 44–55.
- 15 Van den Top, A. M., T. Wensing, M.J.H. Geelen, G. H. Wentink, A. T. van't Klooster, and A. C. Beynen. 1995. Time trends of plasma lipids and hepatic triacylglycerol synthesizing enzymes during postpartum fatty liver development in dairy cows with unlimited access to feed during the dry period. *J. Dairy Sci.* 78: 2208–2220.
- 16 Young, J. W. 1977. Gluconeogenesis in cattle: significance and methodology. *J. Dairy Sci.* 60:1–15.